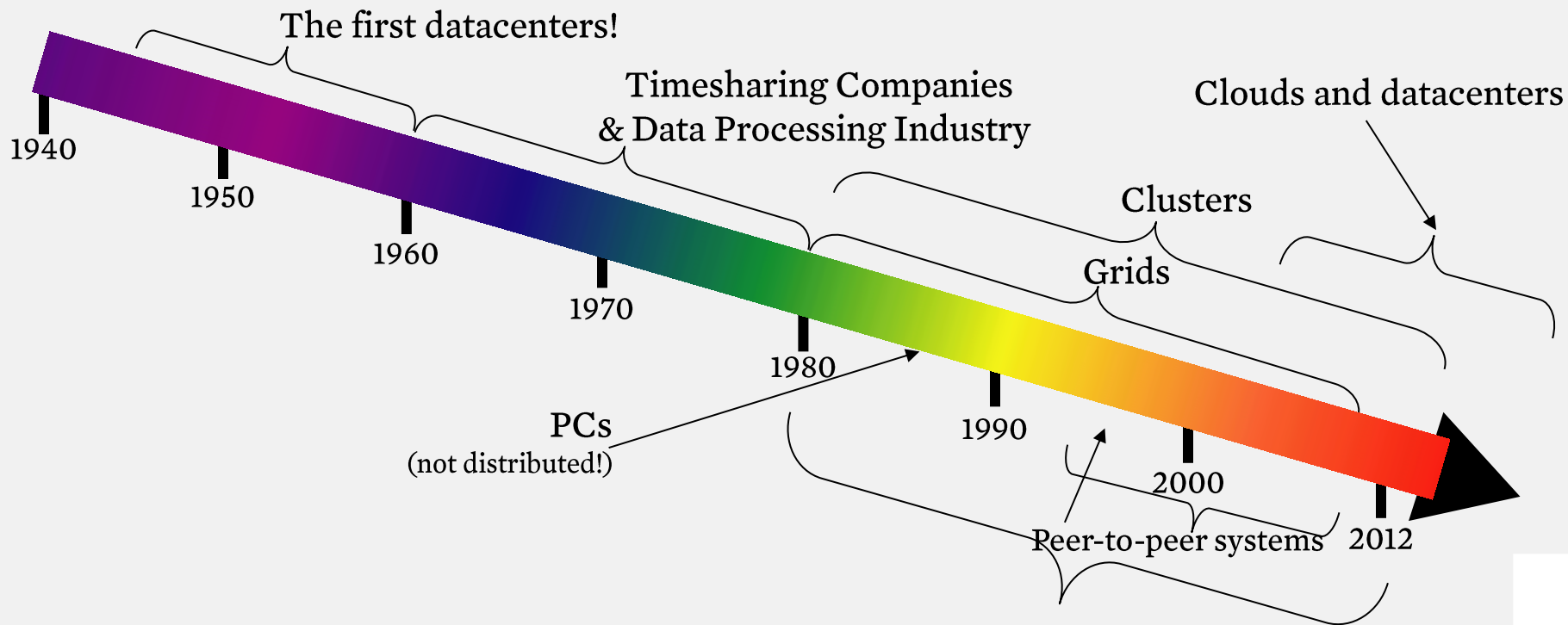
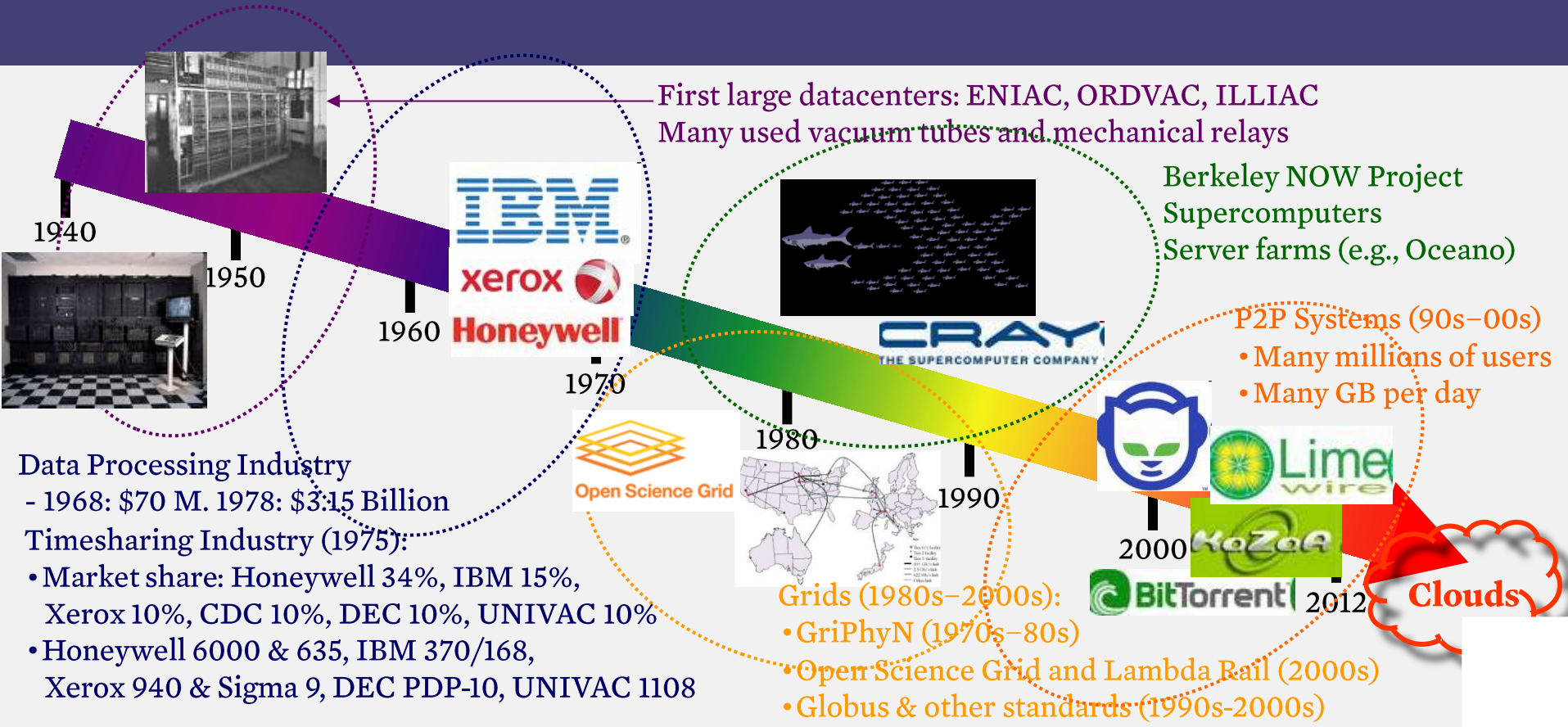


"A CLOUDY HISTORY OF TIME"



"A CLOUDY HISTORY OF TIME"



Key Milestones

- **1960s:** Concept of utility computing introduced.
- **1990s:** Emergence of virtualization technologies enabling multiple operating systems on a single physical machine.
- **2006:** Amazon Web Services (AWS) launched, offering Elastic Compute Cloud (EC2), one of the first scalable cloud platforms.
- **2007 onwards:** Rapid growth of cloud platforms like Google Cloud, Microsoft Azure, and others.

TRENDS: TECHNOLOGY

- Doubling periods – storage: 12 months, bandwidth: 9 months, and (what law is this?) CPU compute capacity: 18 months
- Then and Now
 - Bandwidth
 - 1985: mostly 56Kbps links nationwide
 - 2012: Tbps links widespread
 - Disk capacity
 - Today' s PCs have TBs, far more than a 1990 supercomputer

PROPHECIES

- In 1965, MIT's Fernando Corbató and the other designers of the Multics operating system envisioned a computer facility operating "like a power company or water company."
- **Plug** your thin client into the computing utility **and play** your favorite Intensive Compute & Communicate Application
 - Have today's clouds brought us closer to this reality? Think about it.

FOUR FEATURES NEW IN TODAY'S CLOUDS

I. Massive scale.

II. On-demand access: Pay-as-you-go, no upfront commitment.

- Anyone can access it

III. Data-intensive Nature: What was MBs has now become TBs, PBs and XBs.

- Daily logs, forensics, Web data, etc.
- Humans have data numbness: Wikipedia (large) compress is only about 10 GB!

IV. New Cloud Programming Paradigms: MapReduce/Hadoop, NoSQL/Cassandra/MongoDB and many others.

- High in accessibility and ease of programmability
- Lots of open-source

Combination of one or more of these gives rise to novel and unsolved distributed computing problems in cloud computing.

I. MASSIVE SCALE

- Facebook [GigaOm, 2012]
 - 30K in 2009 -> 60K in 2010 -> 180K in 2012
- Microsoft [NYTimes, 2008]
 - 150K machines
 - Growth rate of 10K per month
 - 80K total running Bing
- Yahoo! [2009]:
 - 100K
 - Split into clusters of 4000
- AWS EC2 [Randy Bias, 2009]
 - 40,000 machines
 - 8 cores/machine
- eBay [2012]: 50K machines
- HP [2012]: 380K in 180 DCs
- Google: A lot

SERVERS



Front

SERVERS



Back

SERVERS



Inside

SERVERS



Some highly secure
(e.g., financial info)

COOLING

- Air sucked in
- Combined with purified water
- Moves cool air through system



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POWER



- $WUE = \text{Annual Water Usage} / \text{IT Equipment Energy (L/kWh)}$
(low is good)
- $PUE = \text{Total Facility Power} / \text{IT Equipment Power}$
(low is good - e.g., Google = 1.11)

Off-site

On-site

TWO CATEGORIES OF CLOUDS

- Can be either a (i) public cloud, or (ii) private cloud
- Private clouds are accessible only to company employees
- Public clouds provide service to any paying customer
- You' re starting a new service/company: should you use a public cloud or purchase your own private cloud?

SINGLE SITE CLOUD: TO OUTSOURCE OR OWN?

- Medium-sized organization: wishes to run a service for M months
 - Service requires 128 servers (1024 cores) and 524 TB
 - Same as UIUC CCT cloud site
- **Outsource** (e.g., via AWS): monthly cost
 - S3 costs: \$0.12 per GB month. EC2 costs: \$0.10 per CPU hour (costs from 2009)
 - Storage = \$ 0.12 X 524 X 1000 ~ \$62 K
 - Total = Storage + CPUs = \$62 K + \$0.10 X 1024 X 24 X 30 ~ \$136 K
- **Own**: monthly cost
 - Storage ~ \$349 K / M
 - Total ~ \$ 1555 K / M + 7.5 K (includes 1 sysadmin / 100 nodes)
 - using 0.45:0.4:0.15 split for hardware:power:network and 3 year lifetime of hardware

SINGLE SITE CLOUD: TO OUTSOURCE OR OWN?

- Breakeven analysis: **more preferable to own if:**

- $\$349 \text{ K} / M < \62 K (storage)
- $\$1555 \text{ K} / M + 7.5 \text{ K} < \136 K (overall)

Breakeven points

- $M > 5.55$ months (storage)
- $M > 12$ months (overall)

- As a result

- **Startups use clouds a lot**
- **Cloud providers benefit monetarily most from storage**

SUMMARY

- Clouds build on many previous generations of distributed systems
- Especially the timesharing and data processing industry of the 1960–70s.
- Need to identify unique aspects of a problem to classify it as a new cloud computing problem
 - Scale, On-demand access, data-intensive, new programming
- Otherwise, the solutions to your problem may already exist!